Name _____ Date _____ Period _____

Lab: Forms of Energy Stations

Station 1: Chemical Energy

1. The bag at this station contains calcium chloride. What does the calcium chloride look like?

   _____
   _____

2. The cup at this station contains water. What does the water look and feel like?

   _____
   _____
   _____

3. Pour the cup of water into the bag with the calcium chloride to mix the reactants. Be sure to seal the bag afterward! What does the mixture look and feel like after they are combined? _____

   _____
   _____

4. This is an example of Chemical Energy! Explain how you think this shows chemical energy: _____

   _____
   _____

5. What other form of energy did you encounter in this reaction? _____ How do you know?

   _____
   _____
   _____
   _____

6. Draw a picture of what you observed in the box below. Be sure to label all items in your picture and make it clear where you see chemical energy.:

Station 2: Nuclear Energy

1. Complete the short reading about nuclear energy at this station and answer the following questions:
2. What is fission?

   _____
   _____
   _____
   _____

3. What element is used to create nuclear energy?

   _____

   _____

   _____

4. What besides generating electricity is nuclear energy used for?

   _____

   _____

   _____

5. What is one advantage and one disadvantage of using nuclear energy?

   _____

   _____

   _____

   _____

Station 3: Gravitational Potential Energy

1. Hold the tennis ball a few inches out from line A. The ball now has Gravitational Potential Energy! Drop the ball and let it bounce a few times. What do you notice happens to the ball each time it bounces?

   _____

   _____

2. Hold the tennis ball a few inches out from line B and drop it and let it bounce a few times. What do you notice is different when you drop the ball from line B compared to line A? _____

   _____

3. Hold the bouncy ball a few inches out from line A and drop it and let it bounce a few times. What do you notice is different from when you dropped the tennis ball from line A? _____

   _____

4. Hold the bouncy ball a few inches out from line B and drop it and let it bounce a few times. What do you notice?

   _____

5. Draw a picture to illustrate what happens at each point in the boxes below. Be sure to label all items in your picture and make it clear where you see gravitational potential energy.:

Tennis Ball:	Bouncy Ball:

Station 4: Electrical Energy

1. Connect the circuit as shown in the picture at the station. What happens?

   _____

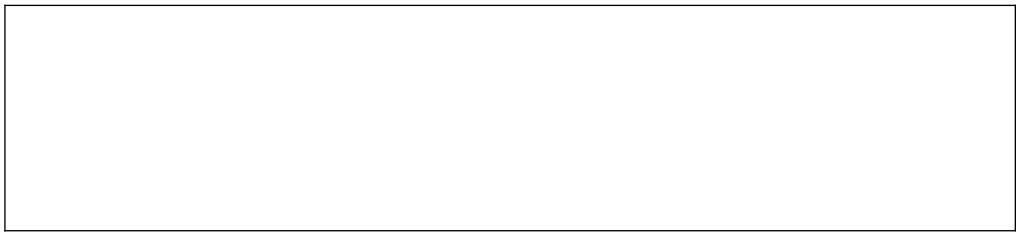
_____

_____

2. Where is the electrical energy coming from?

   _____

3. This is electrical energy! Draw a picture of what you observe in the box below. Be sure to label all items in your picture and make it clear where you see electrical energy.:



4. Disconnect the blue wire from the battery. What happens?

   _____

   _____

   _____

5. Why do you think this happens?

   _____

   _____

   _____

Station 5: Motion Energy

1. Choose one person from your group to spin the coins and one person from your group to be the timer. Have the spinner carefully spin the quarter around on the desk. This is motion energy! Record the time that the coin spins: _____seconds
2. Have the spinner carefully spin the dime around on the desk. Record the time that the dime spins: _____ seconds.
3. What do you notice about the time that the quarter spins and the time that the dime spins? _____

   _____

   _____

4. Why do you think that might be?

   _____

   _____

   _____

5. Draw a picture of what you observed at this station in the box below. Be sure to label all items in your picture and make it clear where you see motion energy.

Station 6: Sound Energy

1. Hold the end of the tuning fork in your hand.
2. Carefully strike the opposite end of the fork with the mallet. The fork is producing sound energy! What do you hear when you hold the fork about an arm's length from your head? _____

   _____

   _____
3. Slowly move the tuning fork towards your head. What do you notice?

   _____

   _____

   _____
4. Strike the tuning fork again and touch it with your hand. What do you notice?

   _____

   _____

   _____
5. Explain what other forms of energy you observe in this experiment.

   _____

   _____

   _____
6. Draw a picture of what you observed at this station in the box below. Be sure to label all items in your picture and make it clear where you see sound energy and any other type of energy.

Station 7: Thermal and Radiant Energy

1. Slowly move the switch up and down a few times and observe what happens to the light: _____

   _____

   _____ This is radiant energy!
2. Place your hand about 2 inches from the bulb and slowly move the switch up and down again. What do you feel?

   _____

   _____

   _____

   _____

   This is thermal energy!
3. Draw a picture of what you observe happening at this station. Be sure to label all items in your picture and make it clear where you see radiant energy and thermal energy.

Additional Questions

1. Which forms of energy are types of